

From Recycling to Second-Life: Assessing Challenges and Policy Gaps in the Electric Vehicle Battery Reverse Supply Chain

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KEYWORDS Reverse supply chain · EV batteries · Second life · Recycling · Circular economy · State-of-Health · Extended Producer Responsibility · Battery passport

01 INTRODUCTION

Road transport electrification is driving a quiet second transition: the retirement of lithium-ion batteries from electric vehicles. Traction packs leave vehicle service at roughly 70-80% of nominal capacity, and end-of-life volumes will rise steeply through the 2030s.

Every retired pack carries three simultaneous identities - hazardous waste, a reservoir of geopolitically concentrated critical materials, and a partially depleted storage asset. Which identity dominates is not decided by electrochemistry.

It is decided by the **reverse supply chain** that collects, transports, tests, grades and routes the pack and by the institutions that govern it.

02 RESEARCH QUESTIONS

RQ1 How is the reverse supply chain for end-of-life EV batteries designed and coordinated, and which operational, informational and inter-organisational challenges constrain it?

RQ2 How is value recovered through reuse, repurposing, remanufacturing and recycling, and how do supply chain factors condition pathway selection?

RQ3 How do policies and business ecosystems shape stakeholder behaviour and reverse supply chain performance across institutional contexts?

Central argument: the binding constraints on EV battery circularity are managerial and institutional, not only technical.

03 METHODOLOGY

Structured narrative review of the reverse supply chain literature for end-of-life EV batteries. Records retrieved from Scopus and Web of Science Core Collection, screened for relevance, and synthesised through a supply chain management lens.

Evidence is organised into three interacting themes and consolidated into a research gap matrix (RG1-RG8), so every proposed direction is anchored in a demonstrated blind spot.

04 CONCEPTUAL FRAMEWORK

THEME 1 · RSC DESIGN & COORDINATION

return uncertainty & ownership · collection channel design
transport & dismantling trade-offs · SoH as an information problem
traceability & battery passport

↓ regulation configures networks & data access

THEME 2 · BATTERY VALUE RECOVERY

reuse → repurpose → remanufacture → recycle
pathway economics & barriers
profitability = chemistry × geography × logistics

↓ incentives, liability & standards shape markets

THEME 3 · POLICY & GOVERNANCE

EPR architecture · EU · China · USA · India
passport & recycled-content rules
circular business models · ecosystem orchestration

Figure 1. Integrated conceptual framework for end-of-life EV battery management from a supply chain management perspective.

05 THEME 1 · RSC DESIGN & COORDINATION

Return uncertainty is multidimensional. Quantity, timing, quality and location of returns vary independently; retirement is a distribution, not a date.

The design lever is disputed. Network-design studies treat uncertainty as exogenous; business-model studies treat it as endogenous to ownership leasing and battery-as-a-service collapse much of it.

Synthesis: ownership design sets the uncertainty regime within which network design operates.

Centralised vs decentralised. Centralised dismantling minimises processing cost; decentralised minimises transport cost and risk. The optimum turns on volume density - an inherently dynamic quantity.

Disassembly is the cornerstone bottleneck. Packs optimised for energy density and crash safety are expensive to reverse; cell-to-pack adhesive designs make it harder.

State-of-Health is an information problem. Usage data sits with OEMs, so buyers discount for uncertainty - depressing valuations and suppressing the second-life market.

06 REVERSE SUPPLY CHAIN PROCESS FLOW

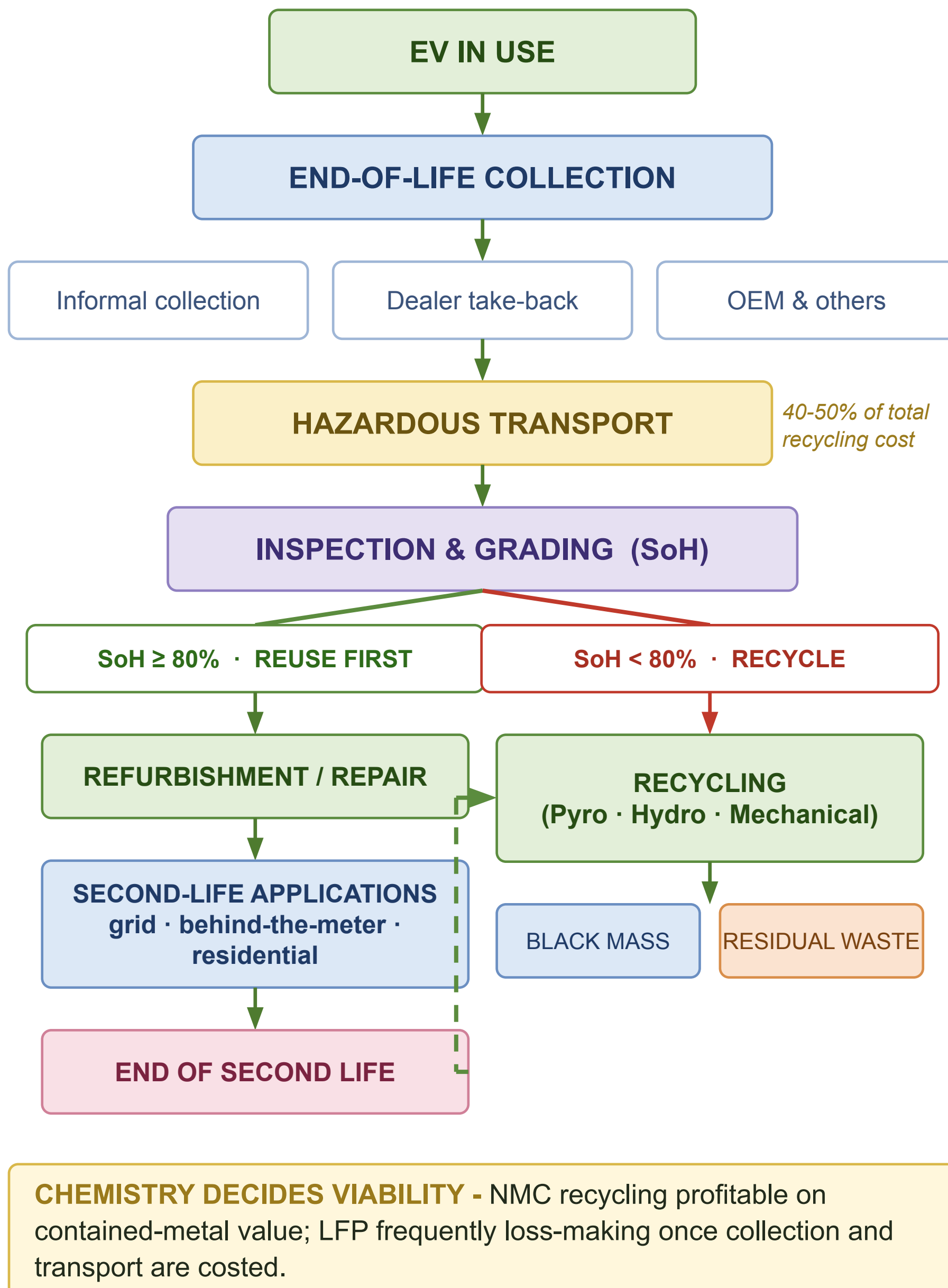


Figure 2. Process flow of end-of-life EV battery management: collection, hazardous transport, State-of-Health grading, and the branching second-life and recycling pathways.

70-80%

residual capacity at pack retirement

40-50%

of recycling cost is transport

2027

EU digital battery passport mandate

RG1- RG8

research gaps identified here

07 THEME 2 · BATTERY VALUE RECOVERY

No single pathway is optimal. Benefit depends on pack condition, the intended second-life application, and the value of recoverable materials.

Profitability is a chemistry × geography × logistics combination, not a property of a technology.

Feedstock timing problem. Manufacturing scrap, not retired packs, dominates near-term recycling feedstock - a capacity-siting challenge as end-of-life volumes arrive later and elsewhere.

Recycling is not unconditionally sustainable. Indian-sector evidence finds recycling can consume more energy and generate more waste than cell manufacture - a further argument for reuse-first routing.

Second life has a migrated bottleneck. Technology (largely resolved) → unit economics (conditionally resolved) → inter-organisational arrangements: who owns, who warrants, who is liable, who sees the data - unresolved and least studied.

08 VALUE-RECOVERY PATHWAYS COMPARED

Pathway	Technical maturity	Economics	Principal SC barriers
Reuse	High for module / cell swap; limited by pack heterogeneity	Attractive where labour cost is low and packs are young; OEM-controlled	Access to BMS data; warranty exposure; repair
Repurposing / second life	Demonstrated in stationary storage pilots and commercial systems	Contested - sensitive to assessment cost, labour and new-battery price decline	SoH information asymmetry; pricing, warranty and liability; standardisation deficits
Remanufacturing	Constrained by non-standard, non-reversible pack designs	Thin evidence base; case-dependent	Design-for-disassembly absent; component supply; certification
Recycling (pyro / hydro / direct)	Industrial scale for pyro & hydro; direct pre-commercial	Chemistry-dependent : profitable for Co/Ni-rich, marginal to negative for LFP	Collection & transport cost; feedstock mix uncertainty; capacity siting

Table 1. Value-recovery pathways compared at the supply chain level.

09 THEME 3 · POLICY & GOVERNANCE

Same principle, different chains. EPR organises policy in every major jurisdiction, yet similar commitments produce dissimilar reverse supply chains - implementation differs on who counts as producer, individual vs collective responsibility, target design and verification.

Regulation is double-edged. Recycled-content mandates create demand certainty for recyclers but can pull packs toward recycling and starve repurposing channels.

Harmonisation, not stringency, is the binding policy variable.

No actor controls enough of the chain. Three unallocated risks - safety liability, warranty exposure, data confidentiality - explain much of the pilot-to-scale gap.

10 COMPARATIVE POLICY ARCHITECTURE

Dimension	European Union	China	United States	India
Core instrument	Regulation (EU) 2023/1542, directly applicable	EPR + traceability platform; interim EoL measures	No federal EPR; state rules and fiscal incentives	Battery Waste Mgmt Rules, 2022 (EPR)
Traceability	Mandatory digital battery passport from 2027	National platform; coding & reporting across life cycle	None mandated federally	EPR registration & reporting; no passport
Material targets	Recovery efficiency + recycled content (Co 16%, Li 6%, Ni 6% by 2031)	Recovery-rate rules for listed recyclers	None federal	Recycled-content & collection obligations phased
Second-life stance	Repurposing recognised; liability rules developing	Cascade utilisation explicitly promoted	Unaddressed federally	Refurbishment recognised within the rules
Key issue	Compliance complexity; passport data governance	Formal-informal leakage; platform coverage	Fragmentation; incentive led circularity	Implementation & informal sector integration unstudied

Table 2. Comparative policy architecture for EV battery end-of-life management.

11 RESEARCH GAP MATRIX & AGENDA

Gap	Future research direction
RG1 · Uneven geographic evidence - networks in China, policy in the EU	Material-flow analysis of formal & informal channels; evaluation of India's 2022 Rules; trade in exported EoL packs
RG2 · Battery health as an information problem, not an estimation problem	Market-design & game-theoretic models of used-battery trade under asymmetric information; pricing studies
RG3 · Second-life contracts - liability, warranty, data access never modelled	Contract-theoretic modelling; case studies of large second-life projects; emerging warranty practice
RG4 · Battery passport in practice ahead of the 2027 mandate	Design & economic analysis of passport data governance; post-2027 evaluation; comparison with China's platform
RG5 · Recycling capacity built for cobalt-rich scrap, not the coming LFP wave	Multi-period capacity & location models under chemistry shift; LFP route costing incl. logistics
RG6 · Policy interactions studied in isolation	Simulation of combined policy effects; battery routing models; evaluation as EU data arrive
RG7 · Absence of management theory despite inter-organisational problems	Transaction cost economics, institutional theory & information-processing views of RSC design
RG8 · Product design and network design studied apart	Combined design-for-disassembly & network optimisation; disassembly costing across pack generations

Table 3. Research gap matrix and future research agenda for supply chain management scholarship.

“The constraints that now bind EV battery circularity are managerial and institutional, not technical.”

- KEY FINDING

12 CONCLUSIONS

1. Return uncertainty can be endogenous by ownership design; information asymmetry relieved by grading economics and data governance; liability ambiguity must be allocated by contract and regulation.

2. **Pathway outcomes are system outcomes.** Whether a pack is reused, repurposed or recycled is decided as much by logistics cost, data access and regulatory architecture as by residual capacity.

3. The literature's strengths are complementary: formal sophistication without empirics in design; qualitative insight without formalisation in second life; EU depth without emerging-market coverage in policy.

Leverage points - practice: data governance & liability allocation · policy: harmonisation over stringency · research: the RG1-RG8 agenda.

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